3/10/26

PRACTICAL - 12

Aim:

To implement echo client server using TCP/UDP sockets.

TCP:

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] connection closed by client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Server received: {data}".encode())
    conn.close()
    server.close()
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] connected to server")
    while True:
        msg = input("Enter message (or 'quit' to exit):")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
```

```
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()
if __name__ == "__main__":
    server_thread = threading.Thread(target = start_server,
daemon = True)
    server_thread.start()
    start_client()
```

UDP :

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_
                                        DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the
connection.")
```

```
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr) # Echo back
    s.close()
    print("[SERVER] Stopped.")
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] closing connection...") break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[CLIENT] Stopped.")
if __name__ == "__main__":
    threading.Thread(target = udp_server, daemon=True).
                start() udp_client()
```

O/P:

```
[server] Received from ('127.0.0.1', 62859): Hell
Echo from server : Hello
Enter message (or 'quit' to stop): quit
```

Result: All

The implement echo client server using TCP Socket executed successfully.